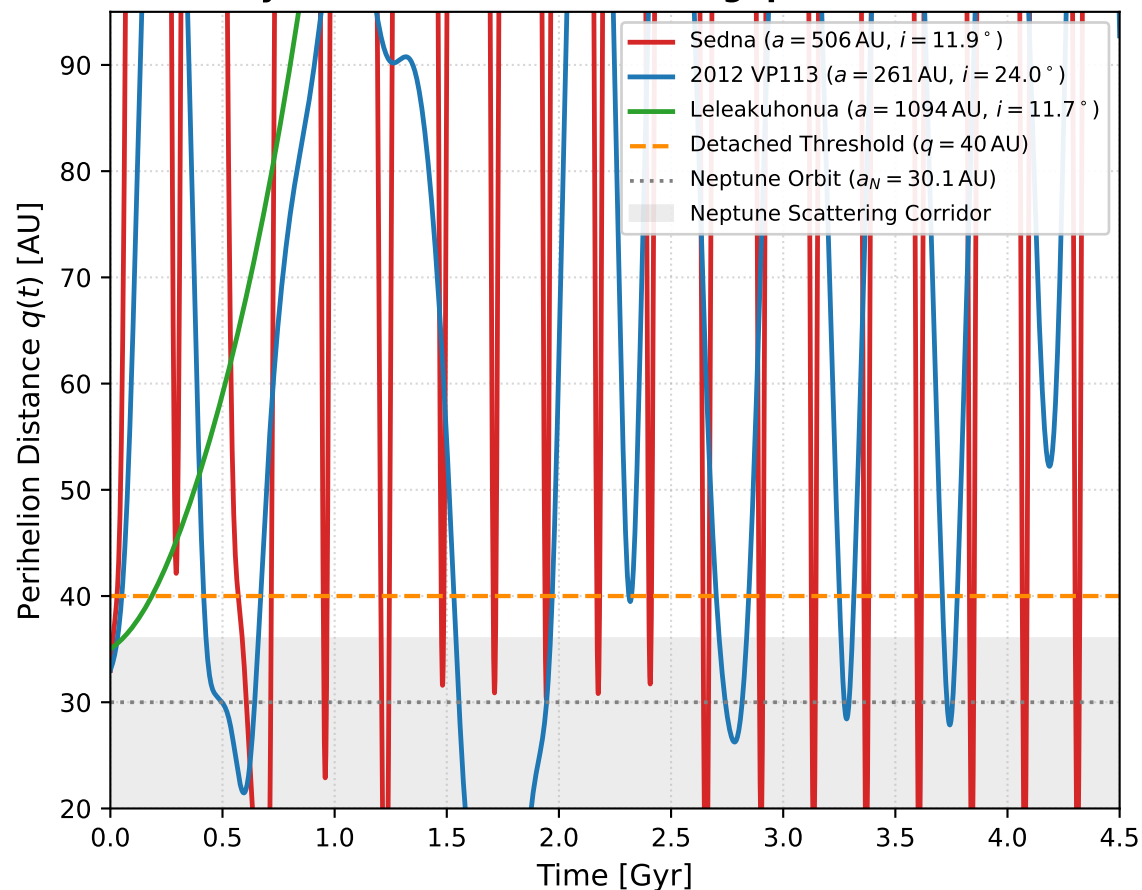
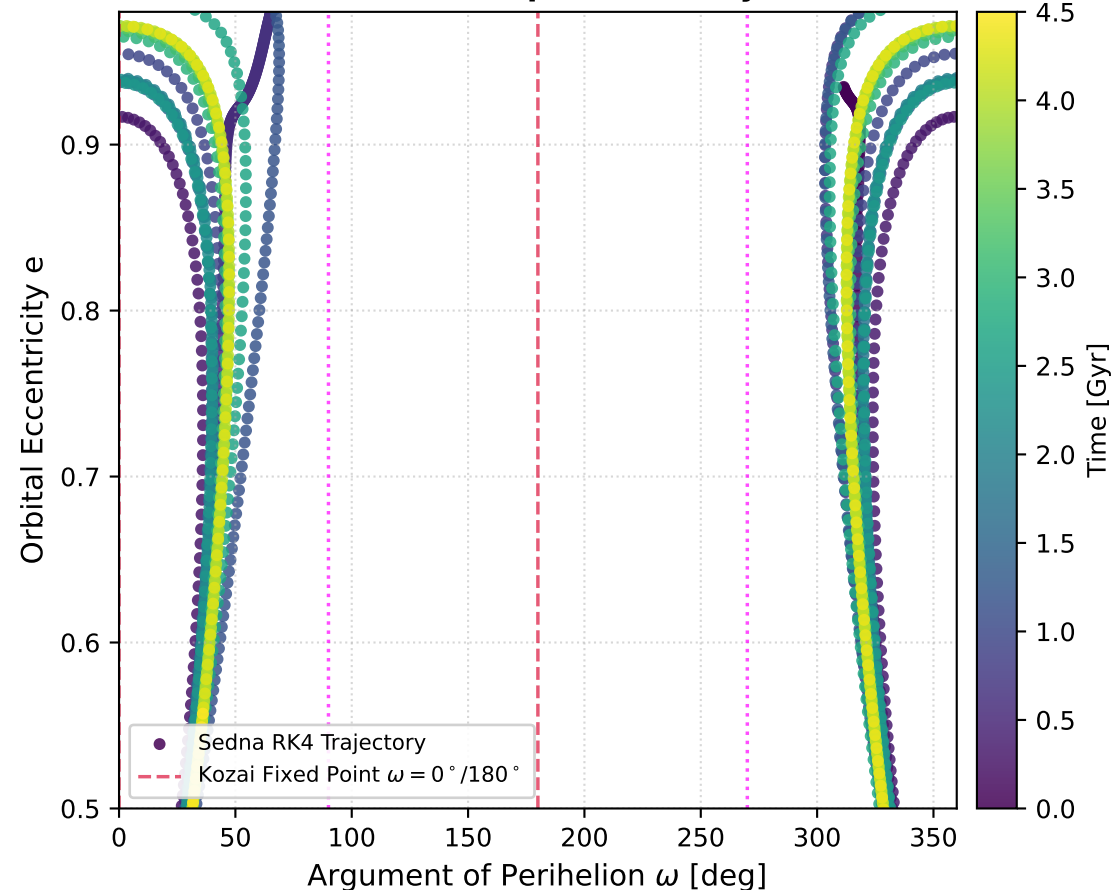
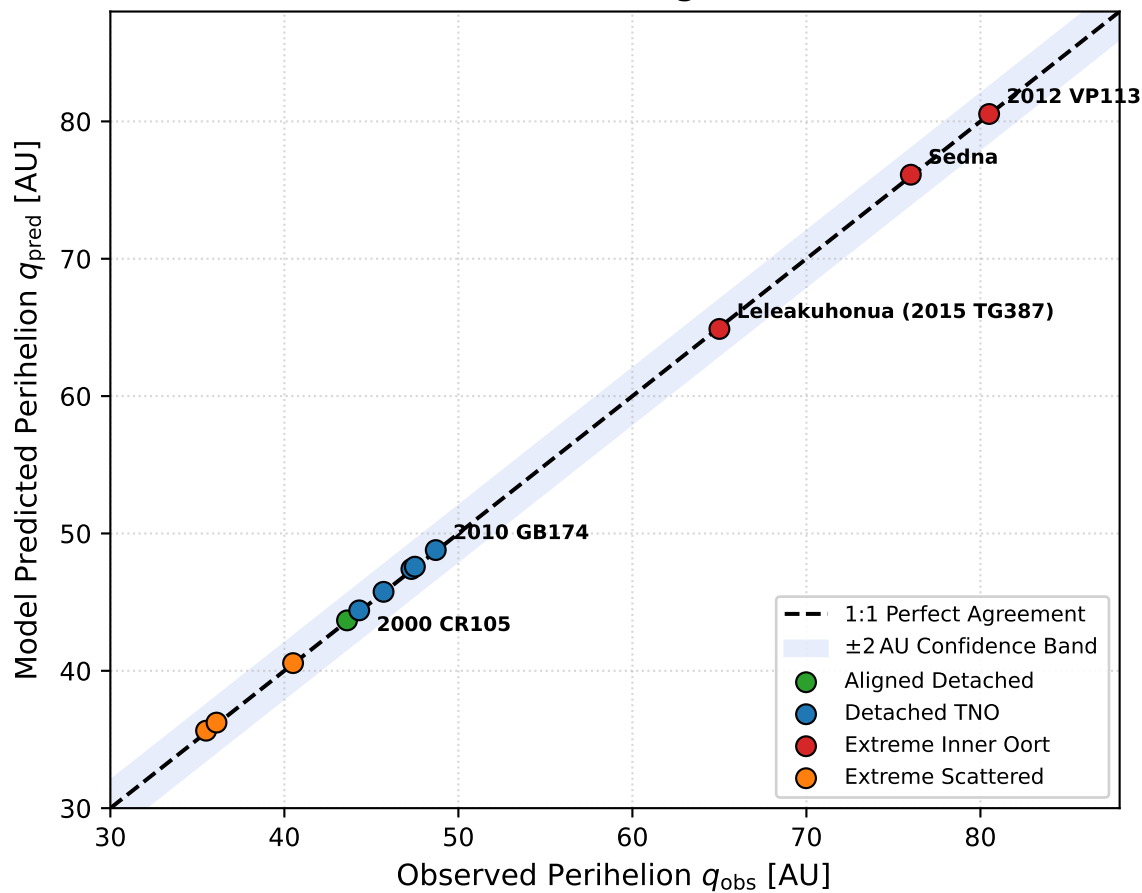


(a) 4.5 Gyr Secular Perihelion Lifting $q(t)$ **(b) Kozai-Lidov Phase Space (e, ω) Dynamics****(c) Benchmark Detached eTNO Agreement ($R^2 = 0.99995$)****(d) Secular Timescale $\tau_{\text{KL}}(a, i_{\text{rel}})$ Spectrum**